



RETAIL SALES ANALYTICS

Executive Report · December 2009 – December 2010

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Data Analytics Portfolio Project
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1 Objective

Analyze 12 months of e-commerce transaction data to quantify **revenue performance**, **customer purchasing behavior and value drivers**, **product contribution**, **geographic concentration**, **cohort retention**, and **RFM-based segmentation** — translating raw transactions into prioritized, actionable business recommendations.

2 Dataset & Methodology

Dimension	Detail
Records (raw → clean)	525,461 → 407,239 transactions
Cleaning applied	Removed 107,927 null Customer IDs, 10,205 cancellations, 3,656 zero-price rows, 425 non-product records (e.g., POST, ADJUST, BANK CHARGES)
Analysis period	1 Dec 2009 – 9 Dec 2010
Techniques	EDA, MoM trend analysis, Pareto analysis, Cohort retention, RFM segmentation (quintile scoring)

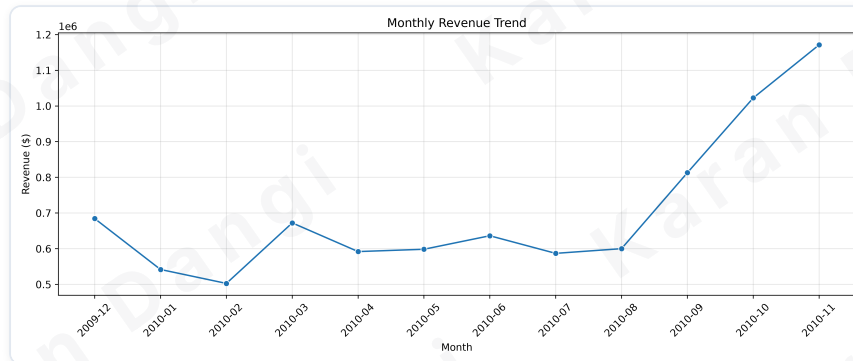
Note: December 2010 contains partial-month data (through 9 Dec 2010), which may understate revenue and retention metrics for the final month.

3 Key Performance Indicators

\$8.73M TOTAL REVENUE	19,098 TOTAL ORDERS	4,286 ACTIVE CUSTOMERS	4,011 UNIQUE PRODUCTS
37 MARKETS (COUNTRIES)	\$457.30 AVG. ORDER VALUE	≈ 60% TOP 10% CUSTOMER REV.	95.19% TOP 5 COUNTRIES REV.

4 Key Findings

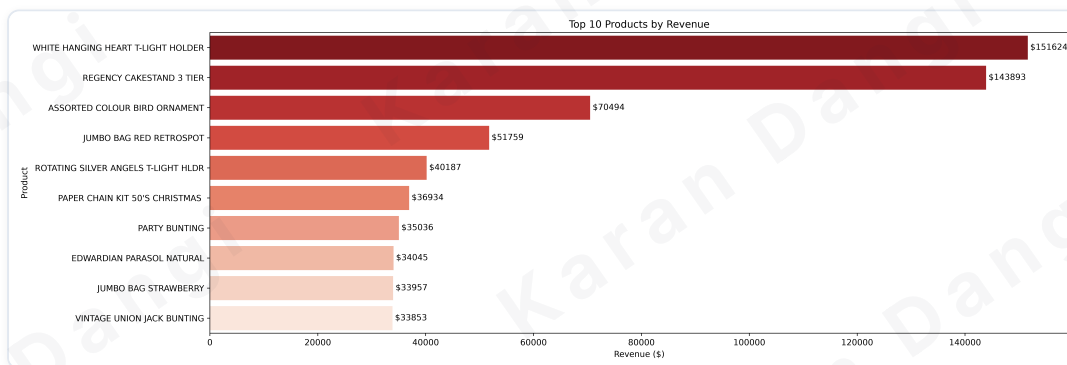
4.1 Revenue Trend — Strong Q4 Acceleration



Quarter	Revenue	MoM Highlights
Q1 (Jan–Mar 2010)	\$1.72M	Jan dip (–20.9%), strong Mar rebound (+33.7%)
Q2 (Apr–Jun 2010)	\$1.83M	Steady recovery, Jun +6.3%
Q3 (Jul–Sep 2010)	\$2.00M	Sep surged +35.5% — early holiday buying
Q4 (Oct–Nov 2010)	\$2.19M	Peak months — Nov reached \$1.17M (highest)

Insight: Revenue increased by approximately **27%** from Q1 to Q4, reflecting increased seasonal demand during the holiday period. The Sep–Nov growth streak (+35.5%, +25.8%, +14.6%) should be considered when planning annual inventory and marketing activities.

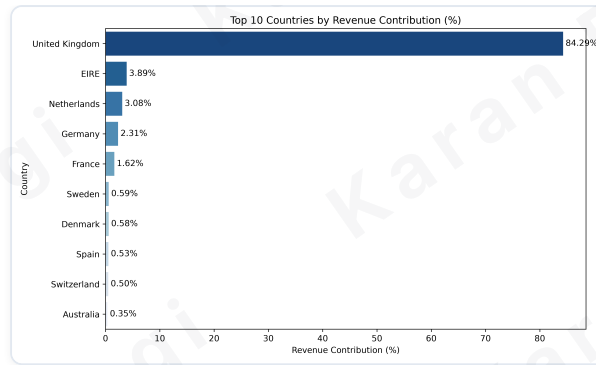
4.2 Product Contribution — Long-Tail Distribution



#	Product	Revenue	Units Sold
1	White Hanging Heart T-Light Holder	\$151,624	56,915
2	Regency Cakestand 3 Tier	\$143,893	12,497
3	Assorted Colour Bird Ornament	\$70,494	44,551

- Top 10 products contribute only **7.28%** of total revenue → long-tail distribution.
- Revenue is broadly distributed across 4,011 products, reducing single-product dependency risk.

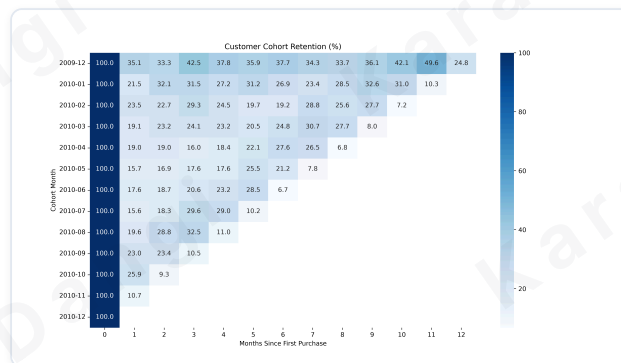
4.3 Geographic Concentration — UK-Dominated



Country	Revenue	Share	AOV
United Kingdom	\$7.36M	84.3%	\$420
EIRE	\$340K	3.9%	\$1,156
Netherlands	\$269K	3.1%	\$2,066
Germany	\$202K	2.3%	\$586
France	\$141K	1.6%	\$606

Insight: Some international markets, particularly EIRE and the Netherlands, exhibit significantly higher AOV than the UK, indicating opportunities for targeted international expansion.

5 Cohort Retention Analysis



Retention Patterns

Metric	Value
Avg. Month-1 retention (2010 cohorts)	~20.5%
Best Month-1 retention	Dec 2009 cohort: 35.1%
Peak single-period retention	Dec 2009, Month 11: 49.6%
Worst Month-1 retention	Nov 2010 cohort: 10.7%

Key observations:

- Approximately **80% of newly acquired customers do not return** in the following month, highlighting customer retention as the largest growth opportunity.
- Dec 2009 cohort shows non-linear re-engagement (retention *increases* at Months 3, 10, 11) — likely driven by seasonal/holiday triggers.
- Later cohorts (Oct–Nov 2010) show steep drop-offs partly due to truncated observation windows.

Opportunity: Even a **5 percentage-point improvement** in Month-1 retention across all cohorts would retain hundreds of additional customers over time and translate to significant incremental revenue.

6 RFM Customer Segmentation

Customers scored on **Recency** (days since last purchase), **Frequency** (unique orders), and **Monetary** (total spend), each on a 1–5 quintile scale. Segments defined using Recency, Frequency and Monetary score combinations.



Segment	Customers	Share	Action Priority
● Need Attention	1,280	29.9%	⚠️ High — personalized re-engagement
● Champions	921	21.5%	🛡️ Protect — loyalty & VIP programs
● Potential Loyalists	656	15.3%	📈 Nurture — convert to loyal
● Lost Customers	636	14.8%	🔄 Win-back campaigns
● Loyal Customers	471	11.0%	🛡️ Retain — exclusive rewards
● At Risk	322	7.5%	🚨 Urgent — prevent churn

Critical insight: 52.2% of the customer base (Need Attention + Lost + At Risk = 2,238 customers) should be prioritized for targeted engagement to reduce churn.

7 Business Recommendations

#	Strategy	Target Segment	Potential Impact
1	Post-purchase drip campaigns within 7–14 days of first order	Need Attention (1,280)	Potential improvement in Month-1 retention
2	VIP loyalty program with tiered rewards & early access	Champions (921) + Loyal (471)	Potential increase in customer lifetime value and customer retention
3	Win-back email sequences with personalized offers & time-limited discounts	Lost (636) + At Risk (322)	Potential recovery of inactive customers
4	Cross-sell bundles featuring top products (T-Light Holder, Cakestand)	Potential Loyalists (656)	Potential increase in AOV and purchase frequency
5	International market expansion — localized campaigns for EIRE, NL, DE, FR	High-AOV markets	Potential growth in international revenue share
6	Seasonal inventory planning — increased stock for Sep–Nov based on Q4 trends	Operations	Reduced risk of stockouts during peak quarter

8 Key Business Risks

Risk	Detail
• Heavy dependence on the UK market	84.3% of total revenue originates from a single market
• Low Month-1 customer retention	Average first-month retention below 25% across all cohorts
• High revenue concentration among top customers	Top 10% of customers generate ≈60% of total revenue
• Strong seasonal dependence on Q4 demand	Revenue is heavily concentrated in the Oct–Nov holiday period, increasing dependence on seasonal demand

9 Conclusion

Finding	Business Implication
Revenue grew ~27% from Q1 to Q4	Seasonal demand should inform inventory and marketing planning
60% of revenue from 10% of customers	Prioritize high-value customer retention
~80% of new customers don't return next month	Improving customer retention represents the largest growth opportunity
52% of customers would benefit from targeted engagement	Segment-specific campaigns can improve retention and support future revenue growth
Some international markets exhibit 2–5× higher AOV than the UK	Targeted expansion offers strong growth potential

Bottom line: By implementing segment-specific retention strategies, optimizing seasonal inventory, and investing in high-AOV international markets, the business has the potential to **significantly improve long-term revenue growth and customer lifetime value.**

Source: UCI Online Retail II Dataset (December 2009 – December 2010) · 407,239 cleaned transactions · 4,286 customers · 37 countries